

Smart Recipe Ontology (SRO)

Phase 2: Data Acquisition, Research Integration & Population

Project Group:

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Course: Ontology Development Project | June 2026

Project Deployments:

 **GitHub:** github.com/EsraSepik/ontology-design-project

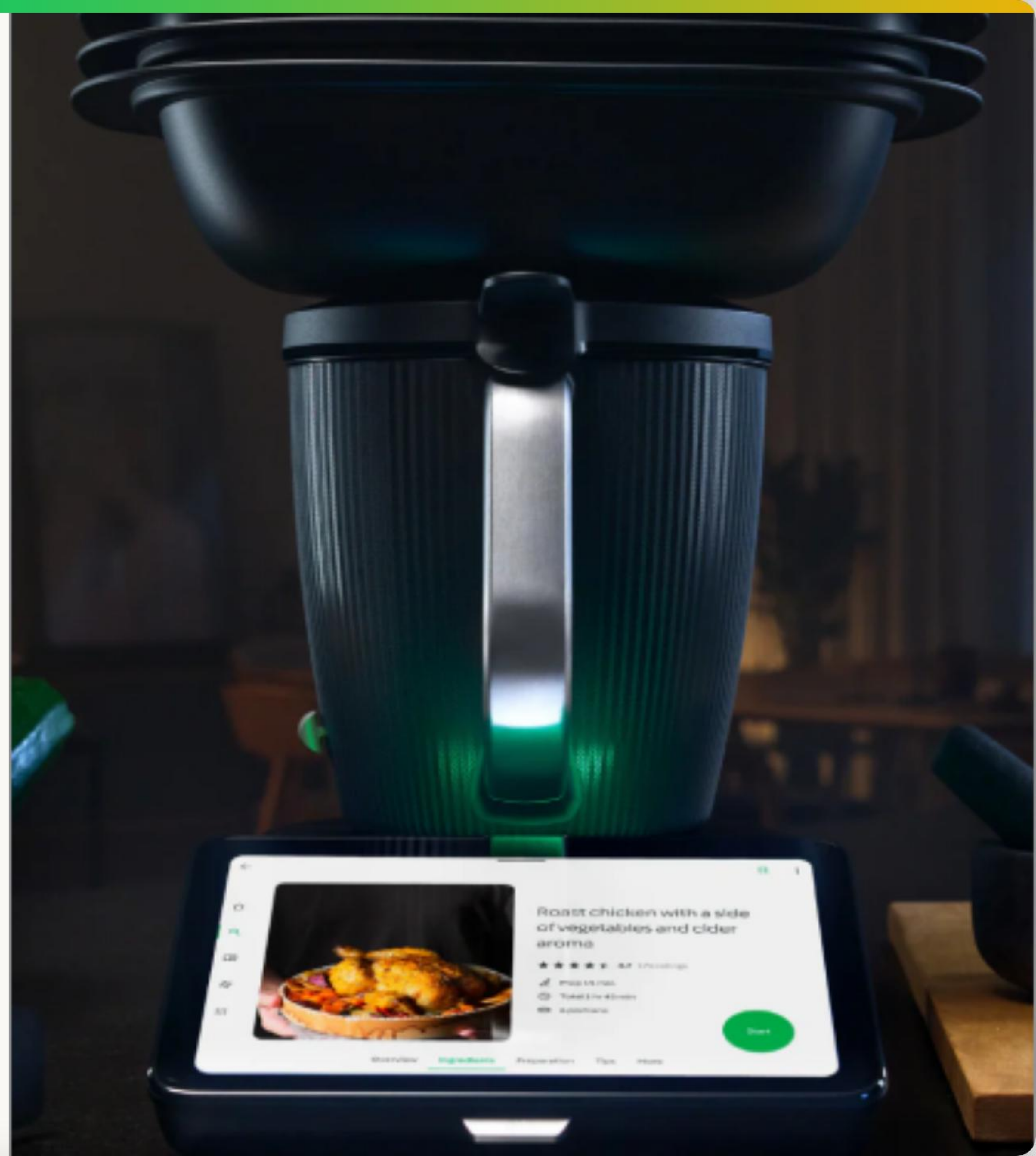
 **WIDOCO:** esrasepik.github.io/ontology-design-project/

Project Mission

Bridging Recipes & AI

SRO provides a machine-readable knowledge model for the culinary world. It transforms unstructured cooking recipes into a structured, standards-compliant Knowledge Graph.

- **Semantic Search:** Easily find recipes by ingredients, diet constraints, and allergen profiles.
- **Automated Ontology Population:** Triples are extracted programmatically using targeted LLM templates.
- **Nutritional and Provenance Tracking:** Every recipe links to QUDT measurements and Dublin Core provenance details.



Evolution: Phase 1 to 2

Feature	Phase 1 (Baseline)	Phase 2 (Extension)
Ontology Scope	Core Recipes & basic Cooking Steps	Nutrition profiles, Provenance metadata & User personalization
Data Entry	Manual individual population	Automated ingestion & extraction pipeline
Validation	Conceptual evaluation	SPARQL, W3C PROV Alignment & strict SHACL Constraints
Research Integration	Basic Competency Questions	LLM populating methods & behavior analysis

Ontology Architecture



Recipe Core

Represents recipes, steps, methods, and cooking tools.

`Recipe, PreparationStep, CookingMethod, Cuisine`



Ingredients

Models ingredient occurrences, quantities, and QUDT units.

`Ingredient, IngredientUsage, Unit, Quantity`



Profiles

Captures allergies, strict diets, and user specifications.

`NutritionProfile, DietType, Allergen, UserPreference`



Provenance

Tracks source information, web links, and validation metadata.

`DataSource, ExtractionConfidence, sourceUrl`

Modeling IngredientUsage

Why separate IngredientUsage from Ingredient?

Standard ontological modeling fails when an ingredient instance (e.g., *Chickpea*) has multiple physical states, decimal quantities, or different metric units across different recipes.

- **N-ary Relation Pattern:** Mediates a robust linkage between a Recipe and an Ingredient alongside contextual properties.
- **Contextual Properties:** Stores food processing states (e.g., "diced", "pre-boiled", "fried") as part of the usage relation.
- **Error Control:** Enables unique provenance metadata and LLM confidence scoring for each ingredient usage occurrence.

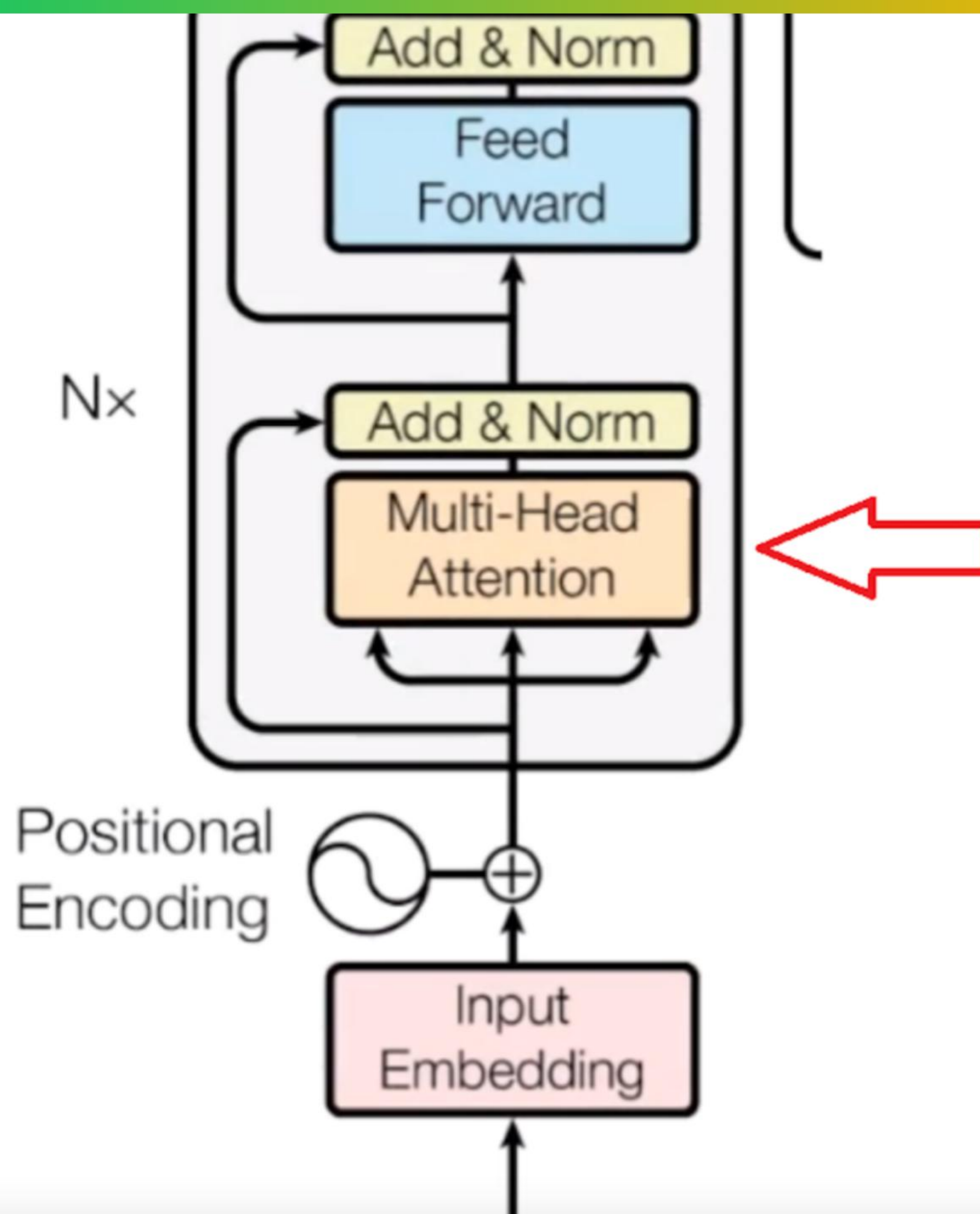


AI Enabler: LLM Ingestion

Ontology-Guided Extraction

Based on the scientific methodology of Norouzi et al., SRO utilizes LLMs to systematically construct RDF graph segments directly from raw recipe text.

- **Schema Constraints:** SRO classes (Recipe, PreparationStep) act as strict prompt guidelines for structural generation.
- **Hallucination Suppression:** Restricting output fields to SRO-defined predicates ensures reliable schema validation.
- **High Efficiency:** Reaches ~90% semantic correctness in triple generation compared to exhaustive manual mapping.



Personalized Experience



Adaptive Filtering

Dynamically matches recipes to distinct calorie budgets, serving sizes, and user preferences using personalized SPARQL queries.



Safety Enforcement

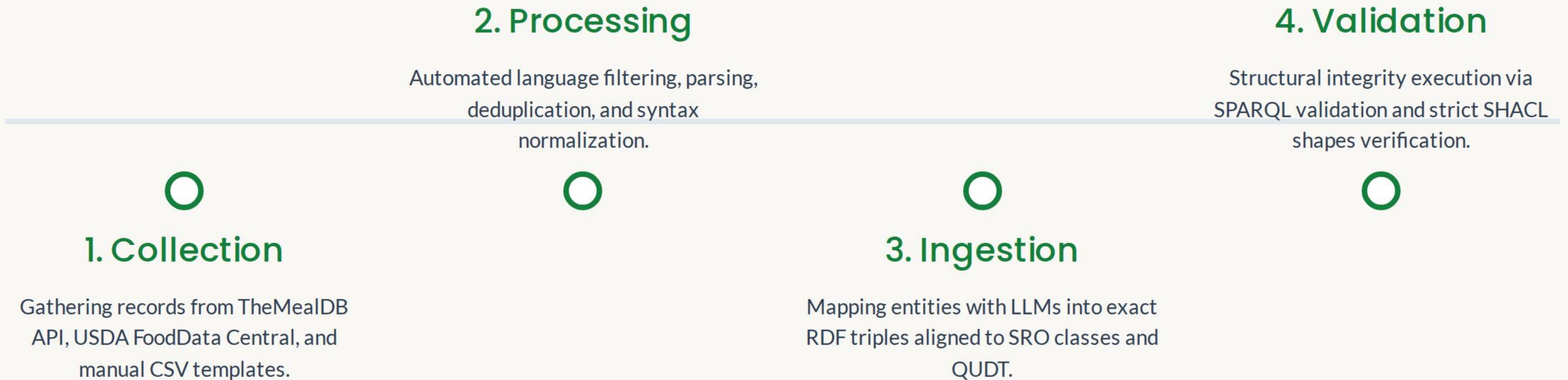
Incorporate strict allergen checking. The ontology automatically filters and flags ingredients registered as allergens in the user profile.



GRU-RNN Foundation

The structured database design lays the semantic foundation for training deep learning models to predict recurring consumer diet patterns.

Data Ingestion Pipeline



Semantic Web Alignments

-  **Schema.org Mapping:** SRO classes `sro:Recipe` and `sro:NutritionProfile` align with standard `schema:Recipe` formats to maximize external search engine compatibility and SEO integration.
-  **QUDT Units Alignment:** Metric values like *grams* or *milliliters* are linked to formal QUDT URIs, ensuring strict mathematical modeling for ingredients.
-  **W3C PROV Provenance:** Tracks dataset origins, extraction timestamps, and machine learning confidence metadata using Dublin Core and W3C PROV vocabularies.
-  **Automated SHACL validation:** Ensures every registered recipe complies with cardinality, decimal boundaries, and expected structural patterns.

Data Quality: SHACL Layer

SHACL Constraints Policy

We enforced validation shapes at `shacl/shapes.ttl` to prevent API anomalies or LLM hallucinations from corrupting the Knowledge Graph.

Key Shapes:

- `sro:RecipeShape`: Mandates title, ingredient list, and steps.
- `sro:PreparationStepShape`: Enforces step numbers are positive.
- `sro:IngredientUsageShape`: Checks for decimals in quantities.

✓ **Overall Result: CONFORMS (True)**

```
# shapes.ttl (Step Validator) sro:PreparationStepShape a
sh:NodeShape ; sh:targetClass sro:PreparationStep ;
sh:property [ sh:path sro:stepNumber ; sh:minCount 1 ;
sh:datatype xsd:integer ; sh:minInclusive 1 ; sh:message
"Positive integer step required" ; ] ; sh:property [ sh:path
sro:instructionText ; sh:minCount 1 ; sh:datatype xsd:string
; ] .
```

SPARQL Competency Queries

Query ID	Competency Question Objective	Validation Result	Status
CQ1	Find vegetarian/vegan recipes based on dietary metadata	Correct recipe individual sets returned	✓ SUCCESS
CQ2 & CQ3	Retrieve quantities and ingredients for a given recipe	All states, units, and values extracted	✓ SUCCESS
CQ4	Nutritional check (recipes below 500 kcal per serving)	Boundary value matches properly evaluated	✓ SUCCESS
CQ8	Filter out gluten/allergens based on user preferences	Sub-graphs filtered safely dynamically	✓ SUCCESS

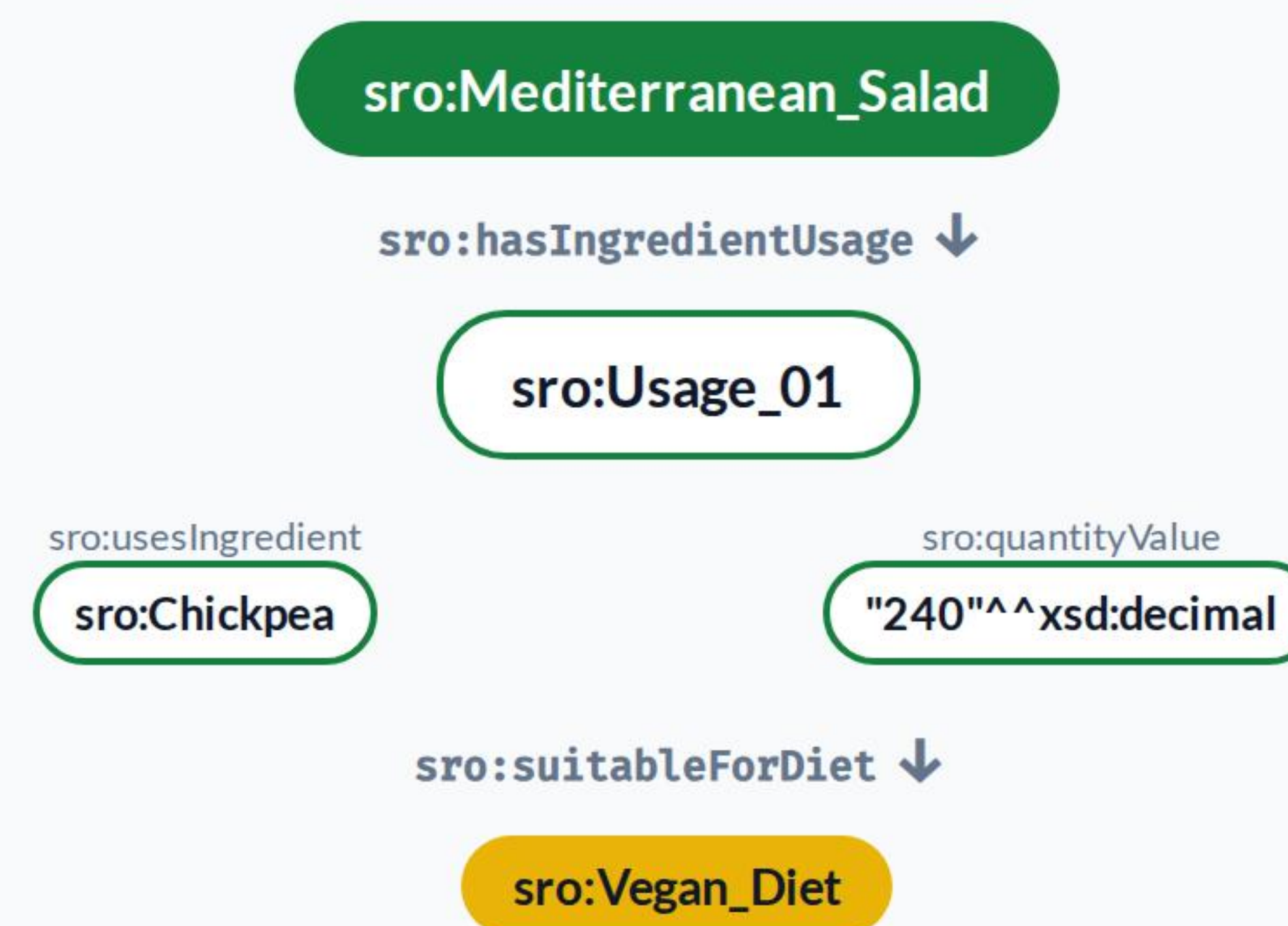
Knowledge Graph Visualized

Graph Relationships

The diagram displays SRO instance relationships modeled for a typical *Mediterranean Salad* recipe.

This structured network lets external search engines, kitchen devices, or dietary chatbots easily fetch recipe allergen profiles, ingredients, and substitutions.

Every individual is semantically grounded in corresponding OWL classes.



Conclusion & Future Work



Achievements

Successfully designed a modular recipe ontology aligned with QUDT standards, implementing a semi-automatic ingestion workflow with rigid SHACL validation.



Dynamic Markets

Future iterations will connect SRO with real-time grocery store APIs, automating pantry tracking and ingredient substitution cost estimation.



AI Behavioral Layer

Deploy deep learning architectures like GRU-RNN directly over populated knowledge graphs to study user dietary patterns and predict custom needs.

Thank You!

Smart Recipe Ontology (SRO) Project | Phase 2 Complete

? Questions & Answers

Please direct queries to Aleyna, Esra, or Sinem.

 GitHub Repository: [EsraSepik/ontology-design-project](https://github.com/EsraSepik/ontology-design-project)

 WIDOCO Site: esrasepik.github.io/ontology-design-project

Image Sources



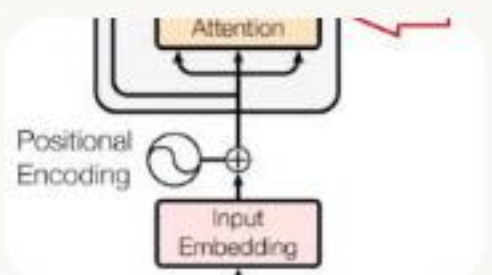
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<https://images.squarespace-cdn.com/content/v1/63064607eb816a4d50027fd1/7f2f9f6a-7659-433d-8617-8e72d94b4b76/mediterranean-diet.jpg>

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